

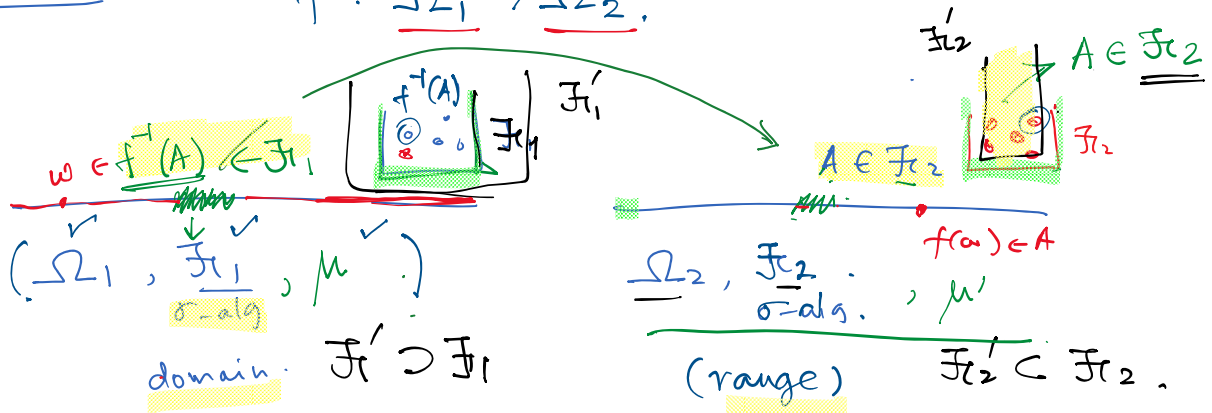
going forward we will only deal with $\mathcal{F} \rightarrow \sigma\text{-algebra}$.

Ch. 2 : random variables & expectations. ✓
(meas. transf.) (Lebesgue integral)

Ch. 2

$$f : \underbrace{\Omega_1}_{\mathcal{F}_1} \rightarrow \underbrace{\Omega_2}_{\mathcal{F}_2}$$

$$f^{-1}(A) = \{\omega \in \Omega_1 : f(\omega) \in A\}$$



f function : $\Omega_1 \rightarrow \Omega_2$

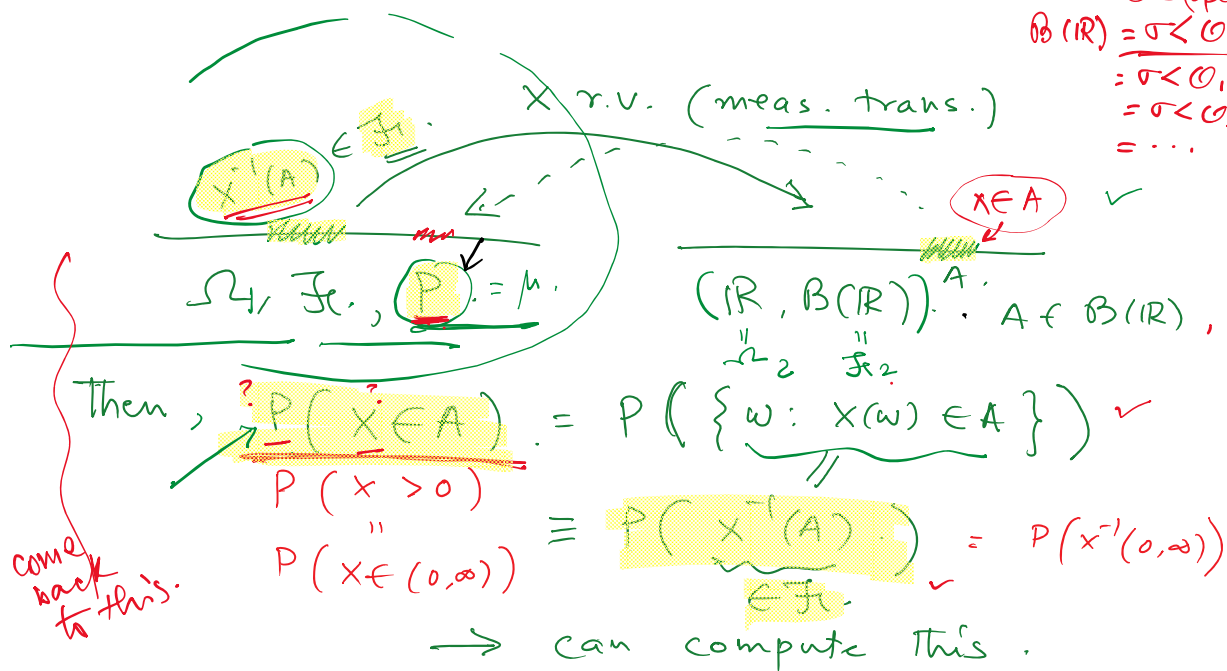
This function f is called measurable (f is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -mble) if:

$$f^{-1}(A) \in \mathcal{F}_1 \quad \forall A \in \mathcal{F}_2$$

Note: If f is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ -mble then it is $\langle \mathcal{F}_1', \mathcal{F}_2' \rangle$ -mble for all $\mathcal{F}_1' \supset \mathcal{F}_1$ and $\mathcal{F}_2' \subset \mathcal{F}_2$.

If left side (domain) has a prob. meas, call it P , and right-side (range) has $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Then a measurable transformation is called a random variable, and is denoted as X, Y etc. (instead of f, g etc.)

$$\begin{aligned} \mathcal{O} &= (\text{open sets}) \\ \mathcal{B}(\mathbb{R}) &= \sigma\langle \mathcal{O} \rangle \\ &= \sigma\langle \mathcal{O}_1 \rangle \\ &= \sigma\langle \mathcal{O}_2 \rangle \\ &= \dots \end{aligned}$$



ex: ① $f: (\Omega_1, \mathcal{F}_1) \rightarrow (\Omega_2, \mathcal{F}_2), c \in \Omega_2$

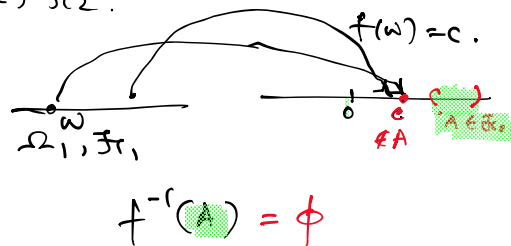
$f(\omega) = c, \forall \omega \in \Omega_1$



take any $A \in \mathcal{F}_2$.

$f^{-1}(A) = \{\omega \in \Omega_1 : f(\omega) \in A\}$

$\dots = \begin{cases} \emptyset & \text{if } c \notin A \\ \Omega_1 & \text{if } c \in A \end{cases}$

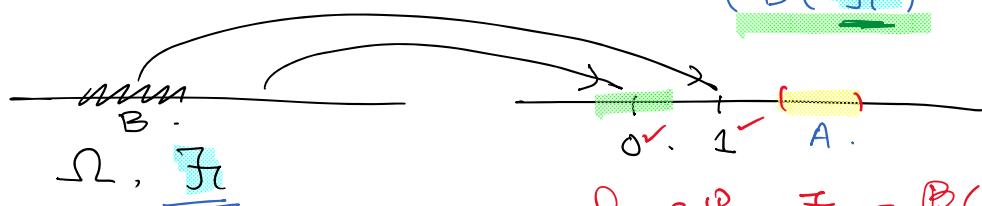


$\in \mathcal{F}_1$ (since $\emptyset, \Omega_1 \in \mathcal{F}_1$).

So constant functions are always meas.

ex: ② $f: (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$

$$f = \underline{1_B}, \quad B \subset \Omega, \quad (B \in \mathcal{F})$$



$$\Omega_2 = \mathbb{R}, \quad \mathcal{F}_2 = \mathcal{B}(\mathbb{R}).$$

$\forall A \in \mathcal{B}(\mathbb{R})$

$$f^{-1}(A) = \{ \omega \in \Omega : f(\omega) \in A \}.$$

$$= \begin{cases} \emptyset \\ B^c \\ B \\ \Omega \end{cases}$$

$$0, 1 \notin A$$

$$0 \in A, 1 \notin A$$

$$0 \notin A, 1 \in A$$

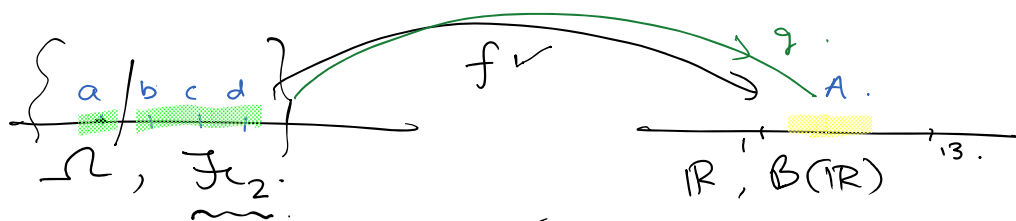
$$0, 1 \in A$$

$$f^{-1}(\{1\}) = B.$$

f is measurable iff $B \in \mathcal{F} \Rightarrow B^c \in \mathcal{F}$ $\mathcal{F}$ σ -alg.

ex ③ mble transformations give "partition" of $\Omega_1 \rightarrow \mathcal{F}_1$ has to be a finer partition than that.

$$\Omega = \{a, b, c, d\}, \quad \mathcal{F} \equiv \mathcal{F}_2 = \{\Omega, \emptyset, \{a\}, \{b, c, d\}\}$$



$$\text{If } f(a) = 1, \quad f(b) = \underline{1/3}, \quad f(c) = \underline{1/3}, \quad f(d) = \underline{1/3}.$$

(check that f is $(\Omega, \mathcal{F}_2)$ mble.

(here $f^{-1}(A) = \emptyset$ or $\Omega \setminus \{a\}$ or $\{b, c, d\}$)

check that $T \sim (\rightarrow, \pi_2)$...

(here $f^{-1}(A) = \emptyset$ or Ω , $\{a\}$ or $\{b, c, d\}$)

$$\begin{aligned} f^{-1}(\{1\}) &= \{\omega \in \Omega : f(\omega) \in \{1\}\} = \{a\} \in \mathcal{F} \\ f^{-1}(\{13\}) &= \{a\}^c \in \mathcal{F} \end{aligned}$$

If $g(a) = 1, g(b) = 2, g(c) = 13, g(d) = -10$

Then g is not $\langle \mathcal{F}_2, \mathcal{B}(\mathbb{R}) \rangle$ mble.

$$g^{-1}(\{13\}) = \{c\} \notin \mathcal{F}_2 \quad \checkmark \quad f^{-1}(\{13\}) = \{a\} \checkmark$$

ex ④ $\Omega = \{H, T\}, \mathcal{F} = \mathcal{P}(\Omega) = \{\emptyset, \Omega, \{H\}, \{T\}\}$

$$\begin{aligned} \checkmark X: \Omega &\rightarrow \mathbb{R} \\ H &\mapsto 1 \\ T &\mapsto 0 \end{aligned} \quad \left\{ \begin{aligned} P(\{H\}) &= p, P(\{T\}) = q. \\ 0 \leq p, q \leq 1, p+q &= 1 \end{aligned} \right.$$

$$\begin{aligned} \Omega, \mathcal{F}, P \text{ prob.} &\xrightarrow{X} \mathbb{R}, \mathcal{B}(\mathbb{R}), \mu(A) = P(X^{-1}(A)) \\ &\quad \uparrow \\ &\quad P_X(A) \end{aligned}$$

$X = 1_{\{H\}}, \{H\} \in \mathcal{F}$ so X is a valid r.v.

$$\begin{aligned} P(X=1) &= P(\{\omega : X(\omega) = 1\}) = P(X^{-1}(\{1\})) \\ &= P(\{H\}) = p \end{aligned}$$

Similarly $P(X=0) = q$

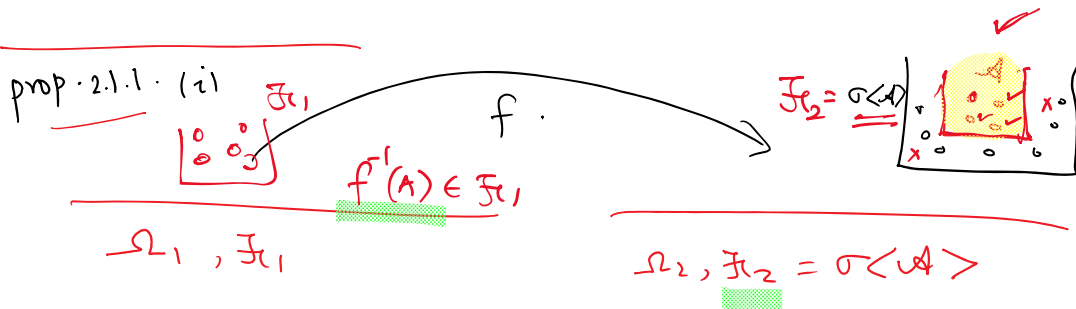
This lets us talk about $P(X \in A) = P(X^{-1}(A))$

$\rightarrow P_X(A) \leftarrow$ a measure on the right side.

This measure is called induced measure (in stat, "distribution of X ") $\rightarrow$ next section,

or "law of r.v."

X

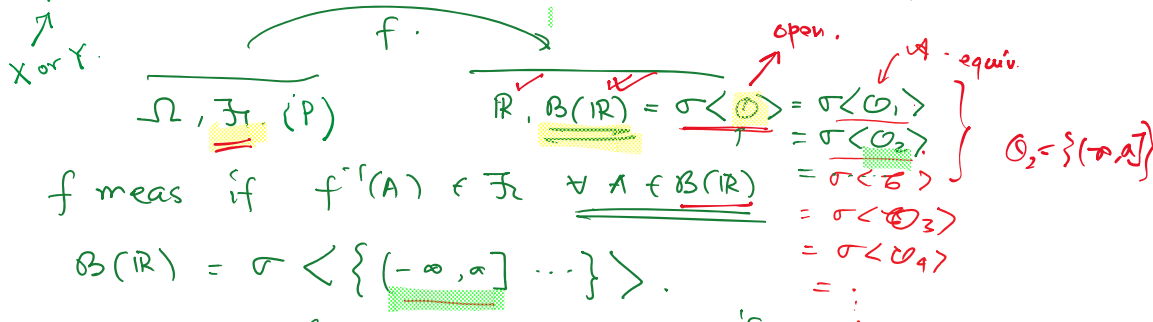


f is measurable ($\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ mble) if

$$f^{-1}(A) \in \mathcal{F}_1 \quad \forall \underline{A \in \mathcal{A}} \cdot (\subseteq \sigma(\mathcal{A}) \underset{\mathcal{F}_2}{\parallel})$$

($\xRightarrow{\text{prop 2.1.1 (i)}}$) $\underline{f^{-1}(A) \in \mathcal{F}_1} \quad \forall A \in \mathcal{F}_2$

f real valued (r.v. if there is a P on $(\Omega, \mathcal{F})$).



f meas if $f^{-1}(A) \in \mathcal{F} \quad \forall A \in \mathcal{B}(\mathbb{R})$

$$\mathcal{B}(\mathbb{R}) = \sigma(\{(-\infty, a] \dots\})$$

f is meas (by prop. 2.1.1 (i)) if

$$\underline{f^{-1}((-\infty, a]) \in \mathcal{F}} \quad \text{for all } a \in \mathbb{R}$$

$$= \{\omega \in \Omega : f(\omega) \in (-\infty, a]\}$$

$$= \{\omega \in \Omega : f(\omega) \leq a\} = \underline{\{f \leq a\}}$$

$\mathcal{B}(\mathbb{R})$ is also $\sigma(\{\text{open sets}\})$,

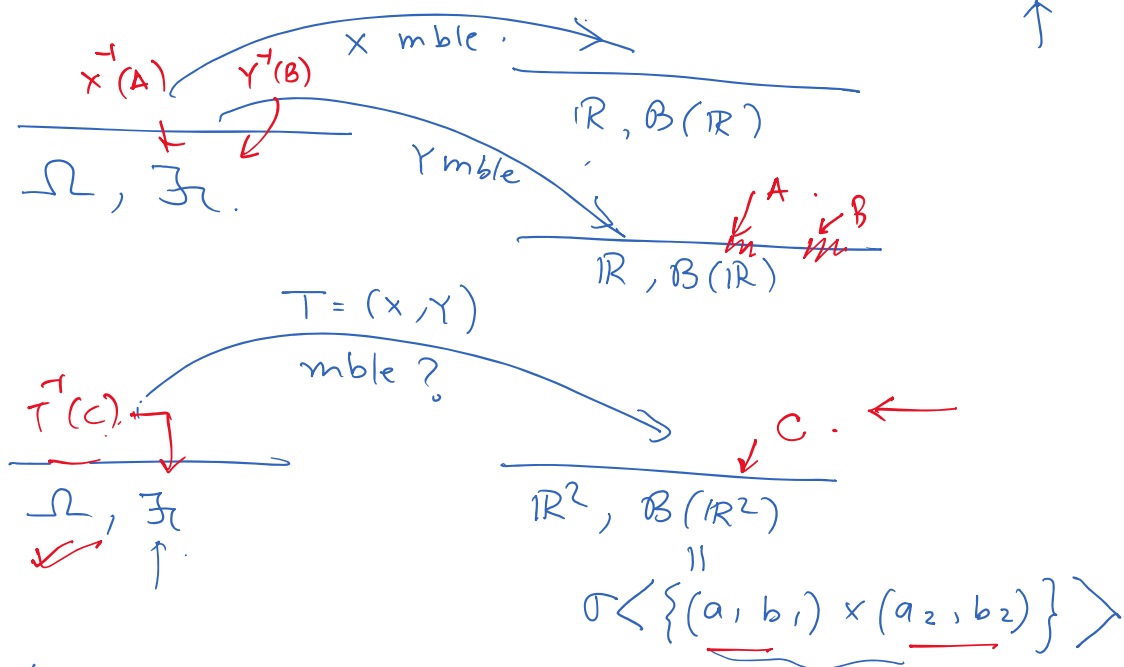
if $f^{-1}(O) \in \mathcal{B}(\mathbb{R})$, for any open set O
Then f is $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ mble.

$f: \mathbb{R}^3 \rightarrow \mathbb{R}^{17}$
 $B(\mathbb{R}) = \sigma\langle \text{open} \rangle \leftarrow B(\mathbb{R}) = \sigma\langle \text{open} \rangle$
 If f cont $f^{-1}(0) = \text{open} \in B(\mathbb{R})$

In particular, if f is cont,
 Then $f^{-1}(0)$ is cont.

i.e. $f^{-1}(0) \in \mathcal{O} \subseteq \sigma\langle \mathcal{O} \rangle \equiv B(\mathbb{R})$
 This shows all cont functions from $\mathbb{R} \rightarrow \mathbb{R}$
 are $\langle B(\mathbb{R}), B(\mathbb{R}) \rangle$ mble.

prop. 2.1.3 (i) : (using X, Y notation
 instead of f_1, f_2 ($k=2$))

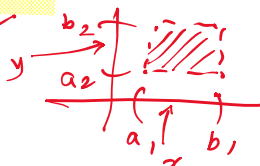


$$T^{-1}((a_1, b_1) \times (a_2, b_2)) = \{\omega \in \Omega : T(\omega) = (X(\omega), Y(\omega)) \in (a_1, b_1) \times (a_2, b_2)\}$$

$$= \{\omega : a_1 < X(\omega) < b_1\} \cap \{\omega : a_2 < Y(\omega) < b_2\}$$

$\in \mathcal{F}$ (since X mble)
 $\in \mathcal{F}$

$$X^{-1}((a_1, b_1)) \in \mathcal{F}$$



$$Z = X + Y = g(T), \quad T: \Omega, \mathcal{F} \rightarrow \mathbb{R}^2, \mathcal{B}(\mathbb{R}^2)$$

$$g(t) = t_1 + t_2 \quad \checkmark$$

Here g is cont. map.
(hence mble)

When $t = (t_1, t_2) \in \mathbb{R}^2$.

$$g: \mathbb{R}^2 \rightarrow \mathbb{R}.$$

and $T = (X, Y)$ is mble.
and comp. of 2 mble maps is mble.

$$\Rightarrow Z = X + Y : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

is mble.

Let $\underline{X}, \underline{Y}$ be measurable, then $Z = \max(X, Y)$

$W = \min(X, Y)$ are meas.

proof: $\{Z \leq a\} = \{\omega: \max(X(\omega), Y(\omega)) \leq a\}$

$Z(\omega) = \max(X(\omega), Y(\omega)) \quad \forall \omega \in \Omega$

alt'ly $W = \min(X, Y)$

$\{W \geq b\} \in \mathcal{F}?$

$$= \{\omega: X(\omega) \leq a\} \cap \{\omega: Y(\omega) \leq a\} \in \mathcal{F}.$$

$\in \mathcal{F} \quad \checkmark \quad \in \mathcal{F} \quad \checkmark \Rightarrow$

$$\min(X, Y) = -\max(-X, -Y) \text{ is meas.}$$

$X, Y \text{ r.v.}$

$$A = \{X = Y\} = \{\omega: X(\omega) = Y(\omega)\}.$$

$$= \{\omega: \underbrace{(X - Y)(\omega)}_{\text{meas.}} = 0\} = (X - Y)^{-1}(\{0\}) \in \mathcal{F}.$$

$(X - Y) \text{ is mble}$

$$B = \{X > Y\} \in \mathcal{F}.$$

$$= \{X - Y > 0\} = (X - Y)^{-1}([0, \infty))$$

We know $\exists A \subset \mathbb{R}$ s.t. $A \notin \mathcal{B}(\mathbb{R})$.

Then $f = 1_A$ is not $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ mble.

$$\mathcal{B}(\overline{\mathbb{R}}) = \sigma \left(\underbrace{\mathcal{B}(\mathbb{R}) \cup \{+\infty\} \cup \{-\infty\}}_{\substack{\text{one elt.} \\ (-a, \infty) \cup \{+\infty\} \\ = (a, \infty]}} \right) \checkmark$$

problem 2.4 (check)

$$f: (\Omega, \mathcal{F}) \rightarrow (\overline{\mathbb{R}}, \mathcal{B}(\overline{\mathbb{R}}))$$

measurable if

$$\forall a \in \mathbb{R}, \{ \omega : f(\omega) \leq a \} \in \mathcal{F} \checkmark$$

$$\left[\underbrace{\{ f = \infty \}}_{\omega \nearrow} = \bigcap_{n \geq 1} \underbrace{\{ f > n \}}_{\in \mathcal{F}} \in \mathcal{F}, \text{ check} \right]$$

$$X: (\Omega, \mathcal{F}) \xrightarrow{\text{mble}} (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

$$\sigma\langle X \rangle = \{ X^{-1}(B) : B \in \mathcal{B}(\mathbb{R}) \} \rightarrow \text{a } \underline{\sigma\text{-alg.}}$$

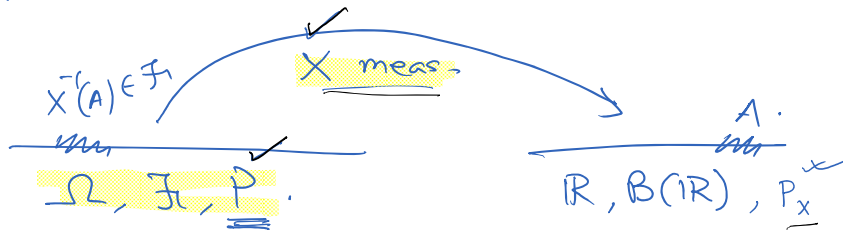
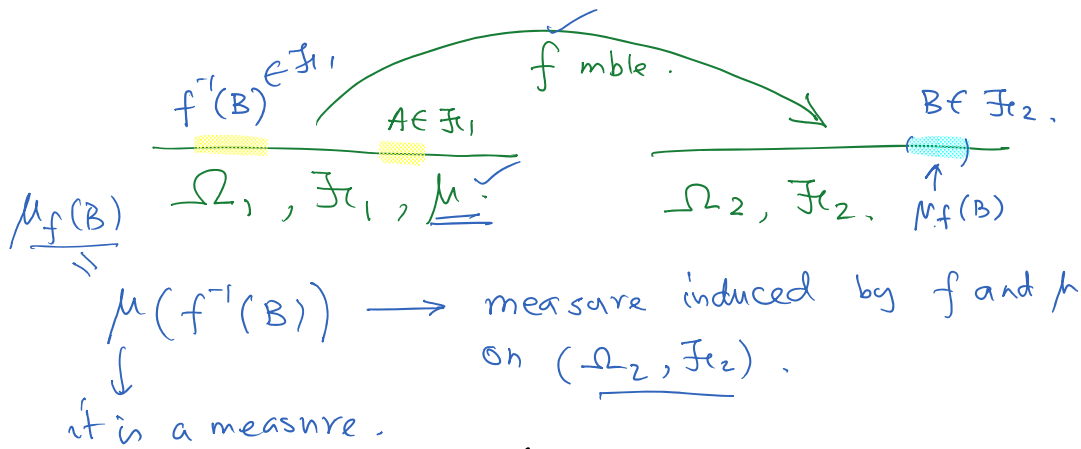
$$\downarrow \subset \mathcal{F}. \text{ (since } X \text{ mble)}$$

σ -alg. gen. by random var. X - (true for other
(smallest σ -alg that makes X mble) meas. fr. too)

2 r.v. X, Y .

$$\sigma\langle X, Y \rangle = \sigma \left(\underbrace{\{ X^{-1}(A) : A \in \mathcal{B}(\mathbb{R}) \} \cup \{ Y^{-1}(B) : B \in \mathcal{B}(\mathbb{R}) \}}_{\text{not a } \sigma\text{-alg.}} \right)$$

Induced measures / distributions :



$P_X(A) = P(X^{-1}(A)) \rightarrow$ is a prob on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$

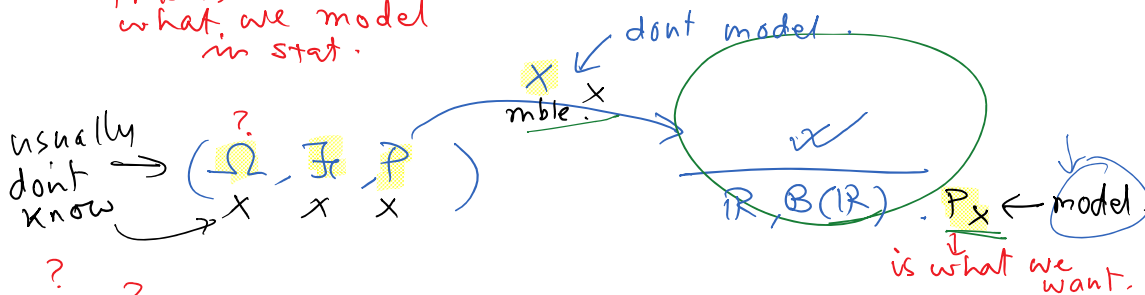
$\hookrightarrow$ distr. of a.r.v. X (on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$)
 (law)

P_X is a meas. on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$
 F_X is a real function.

$X =$ ht of pine trees. $X \sim \text{Gaussian}(\cdot, \cdot)$

P_X is Gaussian measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$

this is what we model in stat.



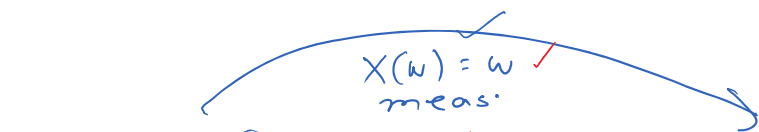
is what we want.

$P(X > 5) = P(X \in (5, \infty)) = P \circ X^{-1}((5, \infty))$
 $= \underline{P_X}((5, \infty))$

Canonical Construction: $\left(\begin{array}{l} \text{Show } \exists (\Omega, \mathcal{F}, P) \text{ and meas. } X \\ \text{s.t. } F_X = F \quad (P_X = \mu_F) \end{array} \right) \left\{ \begin{array}{l} \text{We have a cdf } F \text{ in mind.} \\ (\mu_F) \end{array} \right\}$

$$(\Omega, \mathcal{F}, P) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F)$$

eg. $F = \Phi$
 $\mu_F \rightarrow$ normal measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$



$$\mathbb{R}, \mathcal{B}(\mathbb{R}), P_X = \mu_F$$

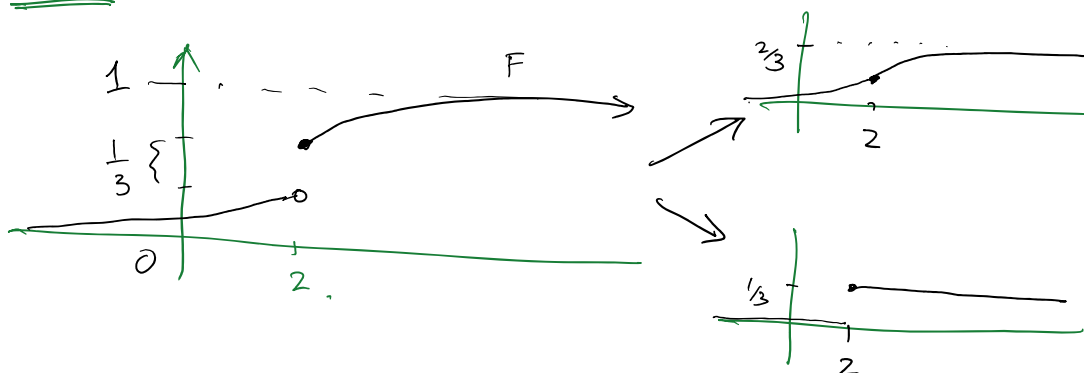
$$P_X(A) = P(X^{-1}(A)) = P(A) = \mu_F(A)$$

$$F_X(x) = P_X((-\infty, x]) = \mu_F((-\infty, x]) = F(x)$$

$$F = \Phi$$

$$\mu_F((a, b]) = \Phi(b) - \Phi(a) = \int_a^b \phi(x) dx \quad | \quad \mu(A) = \dots$$

cdf:



Binomial (n, p) r.v exists?

$$\left[\begin{array}{l} X \text{ is Bin}(n, p) \text{ if } P(X=k) = \binom{n}{k} p^k (1-p)^{n-k} \\ \text{Is there a } (\Omega, \mathcal{F}, P) \text{ and meas. r.v } X \text{ s.t. this holds?} \end{array} \right]$$

Ans: Yes. — use canonical construction.

$$F(x) = \sum_{\substack{k \leq x \\ k \in \{0, 1, \dots, n\}}} \binom{n}{k} p^k (1-p)^{n-k}$$

Can also use other constructions:

$$\Omega = \{ (\omega_1, \dots, \omega_n) : \omega_i = H \text{ or } T, i=1, \dots, n \}$$

$$\mathcal{F} = \mathcal{P}(\Omega)$$

$$P(\{\omega\}) = p^{(\#H)(\omega)} (1-p)^{(\#T)(\omega)} \quad (\#H)(\omega) = \# \text{ of } H\text{'s in } \omega.$$

$$X(\omega) = (\#H)(\omega)$$

$$P(X=k) = (P \circ X^{-1})(\{k\}) = \sum_{\omega \in \Omega, (\#H)(\omega)=k} P(\{\omega\})$$

$$P(X^{-1}(\{k\})) = \binom{n}{k} p^k (1-p)^{n-k} \rightarrow F_X(x) = F_{(X)}^*$$

Third constr.

$$(\Omega, \mathcal{F}, P) = ([0,1], \mathcal{B}([0,1]), m)$$

$$X(\omega) = F^{-1}(\omega)$$

~~Then check that $P_X = \mu$~~

then $F_X(x) = F(x)$.

(We can do that if F has an inverse, otherwise use left / right inverse (see problem 2.15(c)))

— x —

$$F(t) = (1 - e^{-t}) \mathbb{I}_{(t>0)} \leftarrow \text{given to us. (cdf)}$$

$$(\Omega, \mathcal{F}, P) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_F), \quad X(\omega) = \omega. \text{ (mble).}$$

$$P_X(A) = P(X^{-1}(A)) = P(A) = \mu_F(A).$$

$$F_X(x) = P_X((-\infty, x]) = \mu_F((-\infty, x]) = F(x) - F(-\infty) = F(x).$$